

Lecture 11: Government Intervention in the Market

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Lecture overview

- Markets and the Role of Government
- Types of Market Failure
- Government Intervention in the Market
- The Case for Laissez-faire

Markets and the Role of Government

Markets work best when private costs/benefits align with social costs/benefits.
When they differ, we encounter externalities.

- Private vs. Social perspective:
 - Consumer side: We previously assumed only the buyer benefits (Consumer Surplus).
 - Producer side: We assumed all production costs (wages, materials) are borne by the firm (Producer Surplus).

Externalities

The Reality of Externalities:

- Consumption or production often affects third parties not involved in the transaction.
- Consumption Externalities: Private benefit (MPB) $\neq$ Social benefit (MSB).
 - Example: Using a phone while driving increases accident risks for others (Negative externality).
- Production Externalities: Private cost (MPC) $\neq$ Social cost (MSC).
 - Example: A factory polluting a local river imposes costs on the general public.

The Market Failure: In a free market, individuals act in their own self-interest and ignore these external effects.

Objectives of Government Intervention



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The government intervenes to correct market failures, focusing on two goals:

1. Social Efficiency (Allocative Efficiency)

This occurs when resources are allocated such that the net benefit to society is maximized. Economists use the following rules to determine the optimal level of activity:

- If $MSB > MSC$: Society gains more from the last unit than it costs to make; produce/consume more.
- If $MSC > MSB$: The cost to society outweighs the benefit; produce/consume less.

The Optimum ($MSB = MSC$): Resources are perfectly allocated.

Objectives of Government Intervention

2. Equity

- Even if a market is efficient, it may not be fair.
- A free market can lead to extreme inequality (affluence for some, poverty for others).
- Intervention often aims to redistribute resources to achieve a more equitable outcome.

In a perfect market, social efficiency in any activity will be maximised where the activity's:

- A. marginal benefit equals marginal cost.
- B. total benefit equals total cost.
- C. marginal social benefit exceeds marginal social cost.
- D.

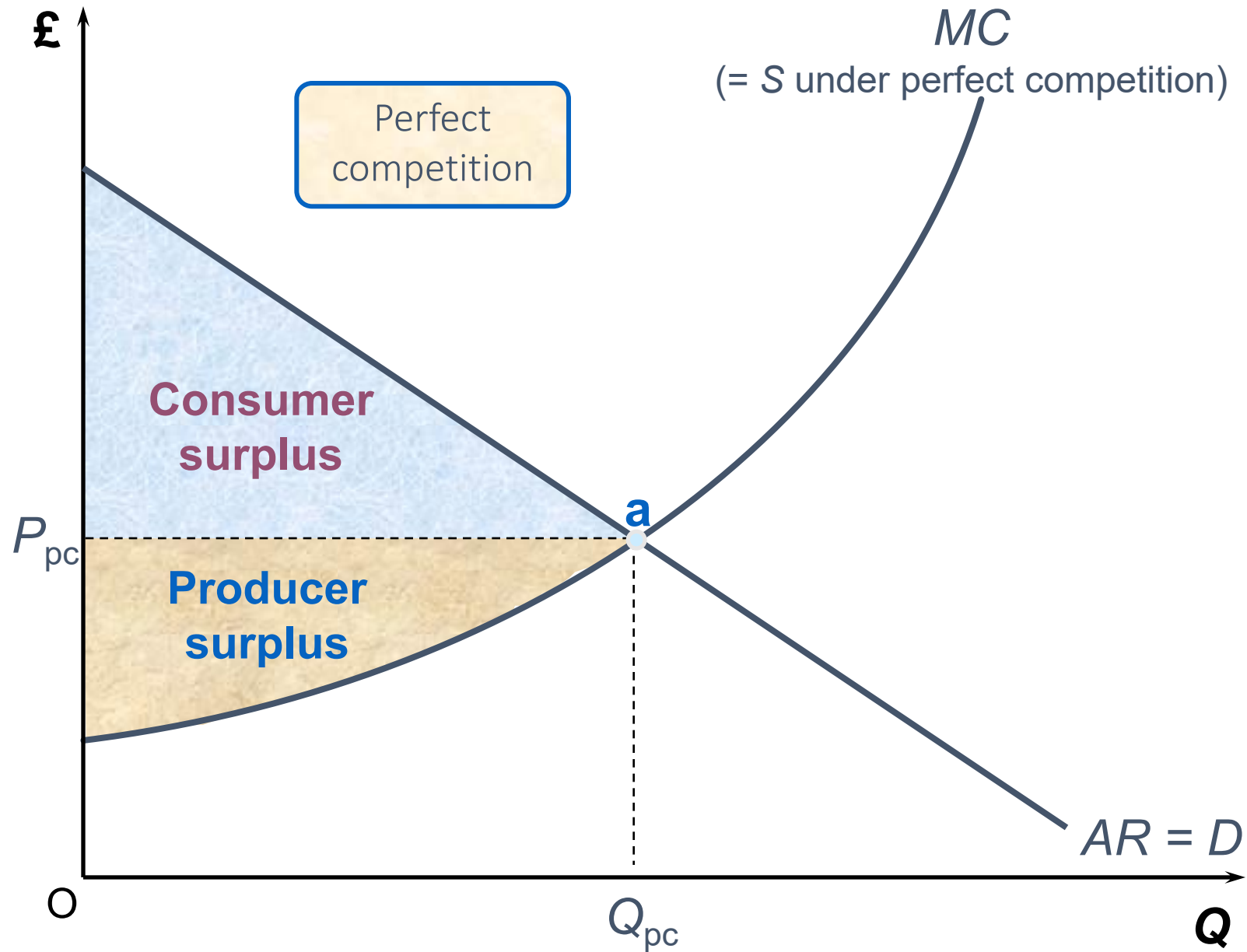
 marginal social benefit equals marginal social cost.
- E. total social benefit exceeds total social cost.

Types of Market Failure

Market Power and Market Failure

- Whenever market power exists (Pure Monopoly, Oligopoly, or Monopolistic Competition), the market fails to reach a socially efficient equilibrium.
- In these structures, firms have the power to set prices above marginal cost, leading to an under-allocation of resources.

Deadweight loss under monopoly

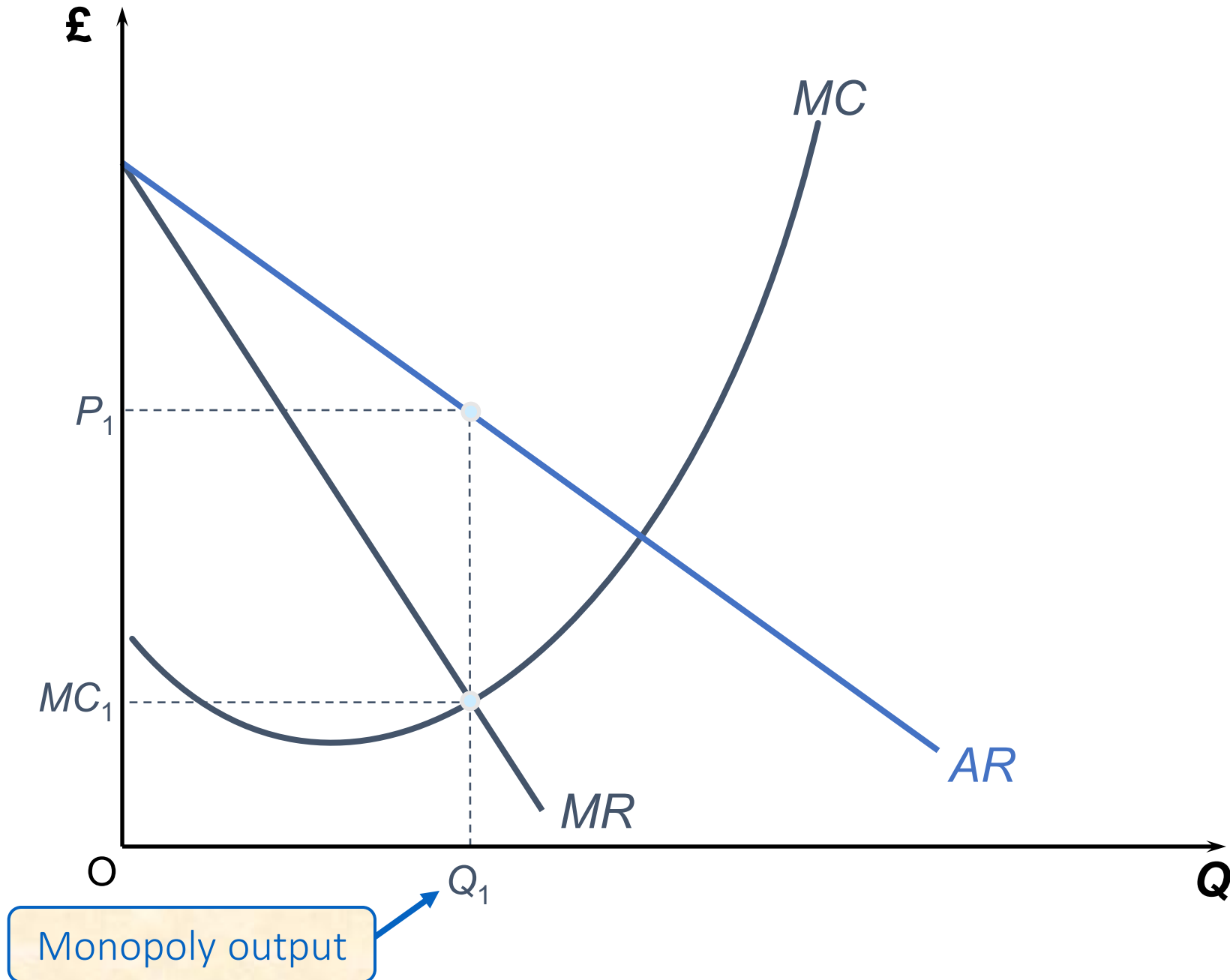


(a) Industry equilibrium under perfect competition

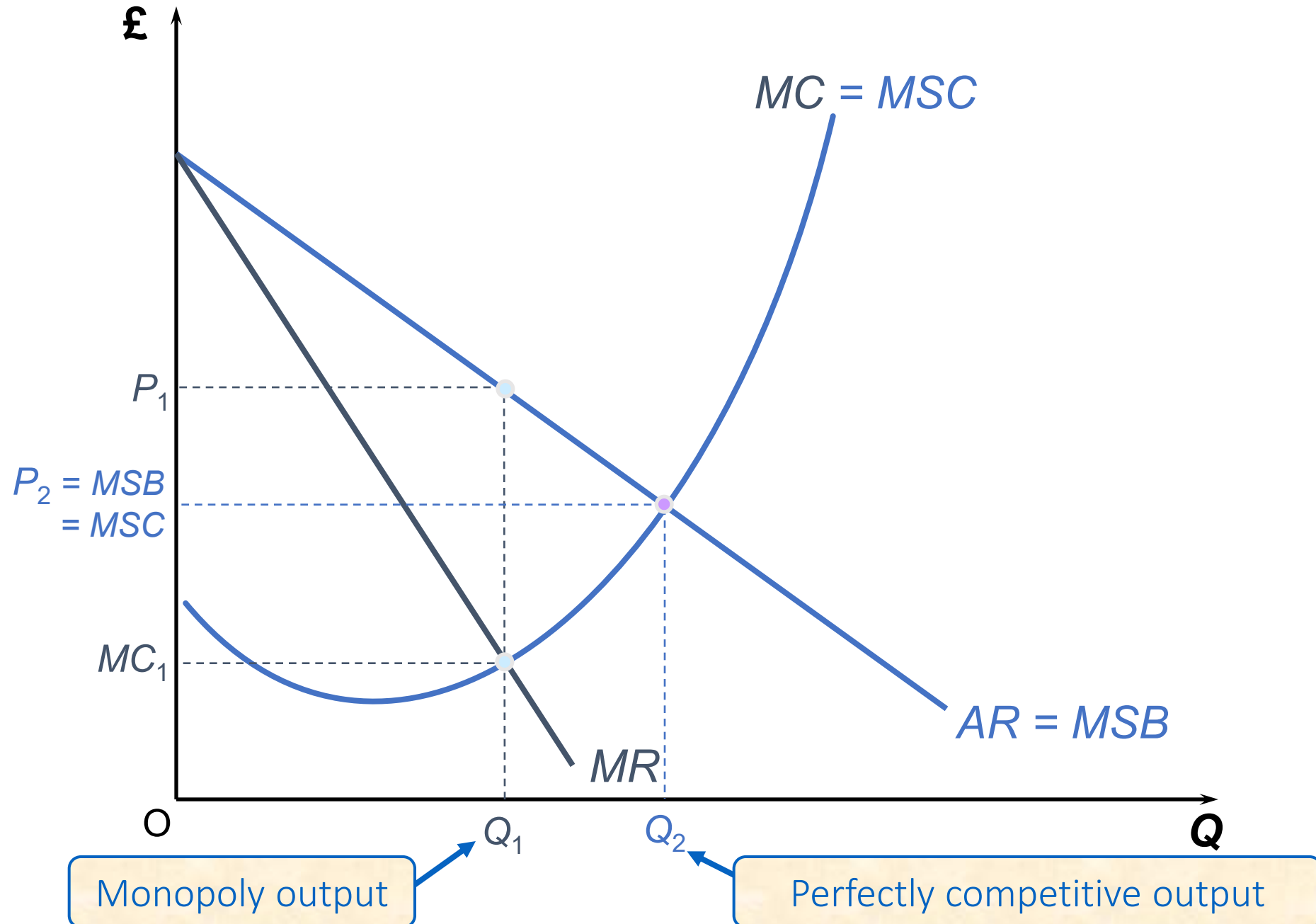
Monopoly and Social Efficiency

- To analyze the impact of market power, we assume for now that there are no external costs or benefits ($MC = MSC$ and $Price/AR = MSB$).
- Profit Maximization vs. Social Efficiency:
 - A monopolist maximizes profit where $MR = MC$ (Output Q_1).
 - Social efficiency requires producing where $MSB = MSC$ (Output Q_2).
- Underproduction: Because a monopolist faces a downward-sloping demand curve, MR is always below AR (Price).
- Consequently, the monopolist produces less than the socially optimal amount ($Q_1 < Q_2$).

A monopolist producing less than the social optimum



A monopolist producing less than the social optimum



Measuring Welfare: Consumer and Producer Surplus

Welfare loss is best understood by comparing Perfect Competition (PC) to Monopoly:

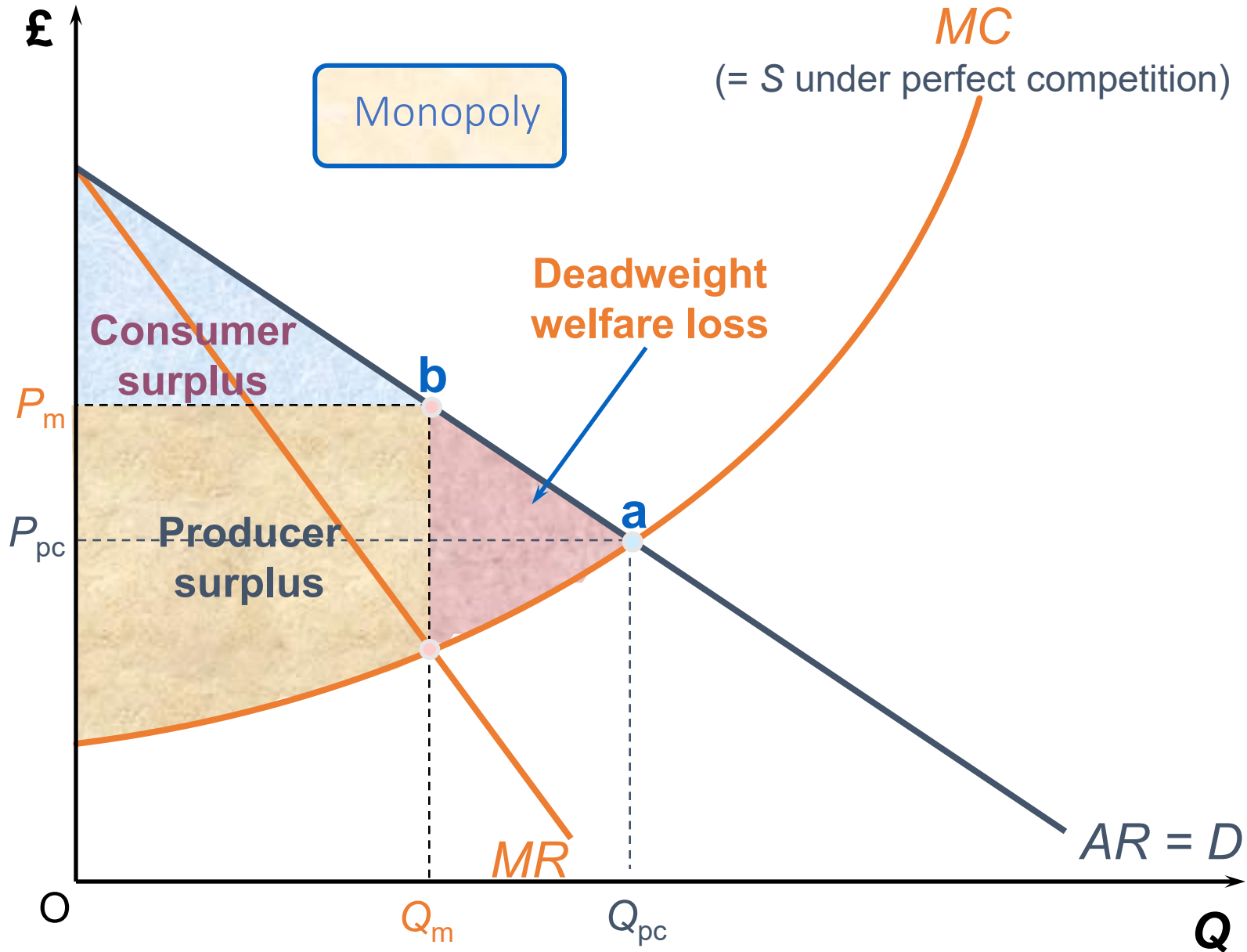
- Consumer Surplus (CS): The difference between what consumers are willing to pay and what they actually pay.
 - In PC, CS is large (the entire area above the price P_{pc} and below the demand curve).
- Producer Surplus (PS): The difference between the price received and the minimum price required to supply (Total Revenue minus Total Variable Cost).
 - In PC, PS is the area above the MC curve and below the price P_{pc} .
- Total Surplus: The sum of CS and PS (the total area between the demand and MC curves).

The Effect of Monopolisation

When an industry moves from Perfect Competition to Monopoly:

- Higher Prices, Lower Output: Price rises from P_{pc} to P_m , and quantity falls from Q_{pc} to Q_m .
- A portion of the consumer surplus is transferred to the producer (as higher profits).
- The gain to the producer is smaller than the loss to the consumer.
- Deadweight Welfare Loss: The combined loss of areas that were previously part of the total surplus under competition but vanish under monopoly. This represents the cost to society of market power.

Deadweight loss under monopoly



(b) Industry equilibrium under monopoly

Which one of the following is a definition of *deadweight welfare loss* under monopoly?

- A. The loss in output compared with perfect competition.
- B. The increase in price compared with perfect competition.
- C. The loss of total consumer surplus compared with perfect competition.
- D. The increase in total profit compared with perfect competition.
- E.

 The loss of total consumer-plus-producer surplus compared with perfect competition.

Terminology

- **MPC** (Marginal Private Cost): extra cost to the producer for one more unit (e.g., labour, raw materials).
- **MEC** (Marginal External Cost): external (third-party) cost created by production/consumption that the firm or consumer does not pay.
- **MSC** (Marginal Social Cost): total extra cost to society = $\text{MPC} + \text{MEC}$.
- **MPB** (Marginal Private Benefit): benefit to the consumer from one more unit (often reflected by demand).
- **MEB** (Marginal External Benefit): external benefit received by third parties from production/consumption.
- **MSB** (Marginal Social Benefit): total benefit to society == $\text{MPB} + \text{MEB}$

Production Externalities

- We examine four types of externalities one by one.
- To analyze the impact of externalities, we isolate specific cases while assuming the rest of the market remains perfectly competitive.

Negative Production Externalities ($MSC > MPC$)

This occurs when a firm's production process imposes a cost on others for which the firm does not pay.

- Example: A chemical plant dumping waste into a river or polluting the air. The community pays for health issues or cleanup, not the firm.
- $MSC = MPC + MEC_p$. Because MEC_p is positive, the MSC curve lies above the MPC curve.

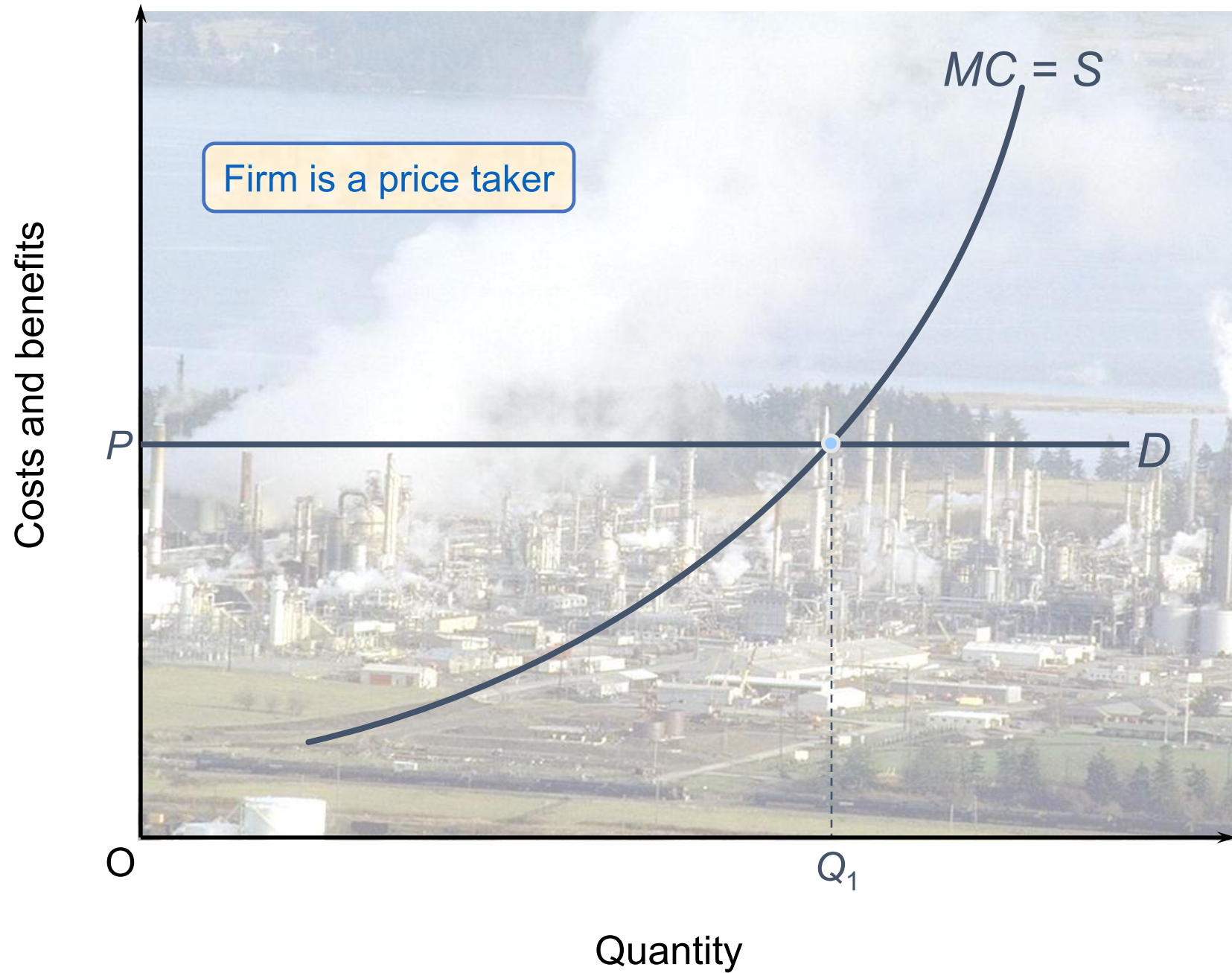
Market Outcome vs. Social Optimum:

- The Free Market: Produces at Q_1 (where Demand/ $MPB = MPC$).
- Social Efficiency: Occurs at Q_2 (where $MSB = MSC$).

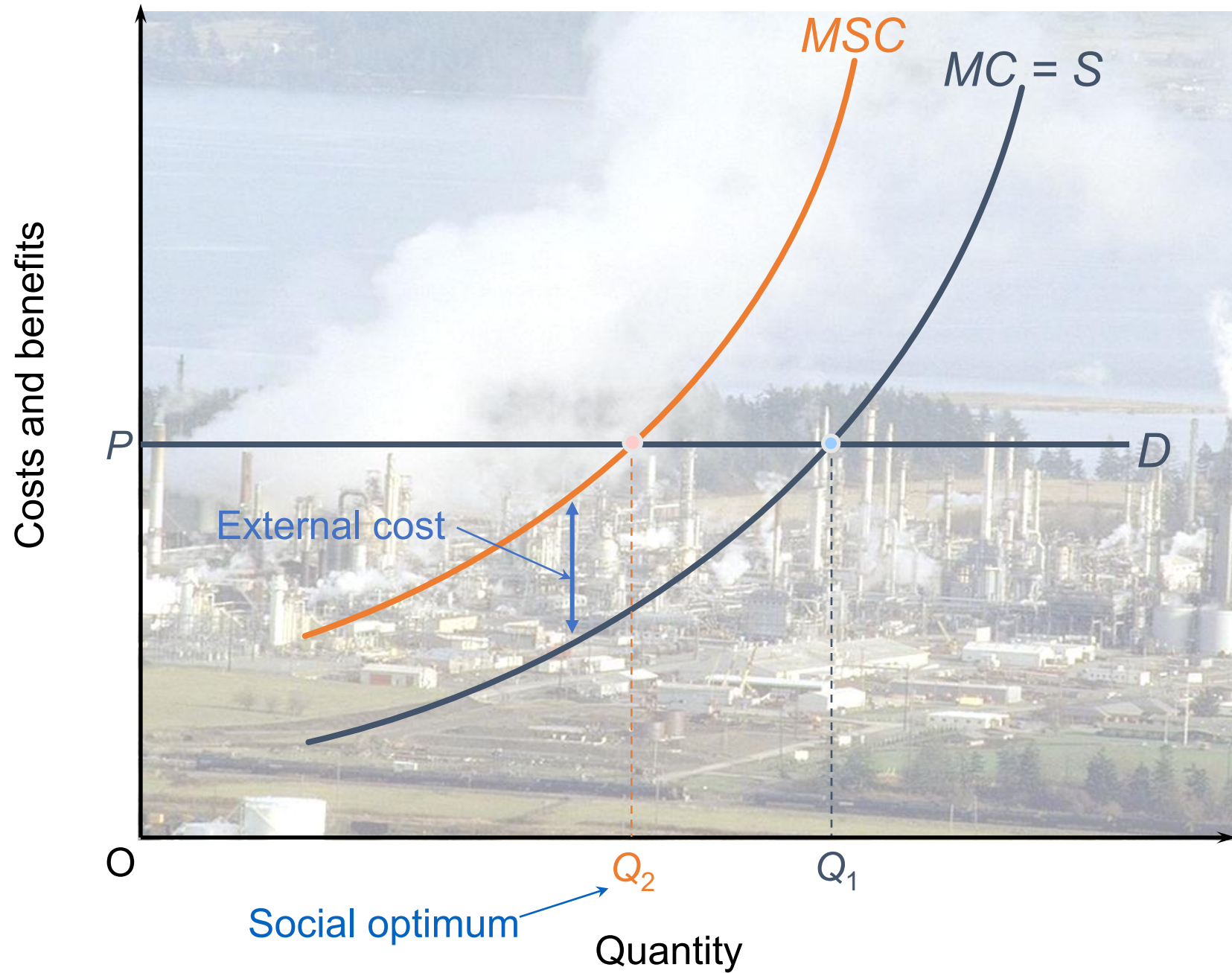
The market produces more than is socially desirable ($Q_1 - Q_2$).

A missing market, no one owns the air or river, so firms use them as free dumping grounds.

External costs in production



External costs in production



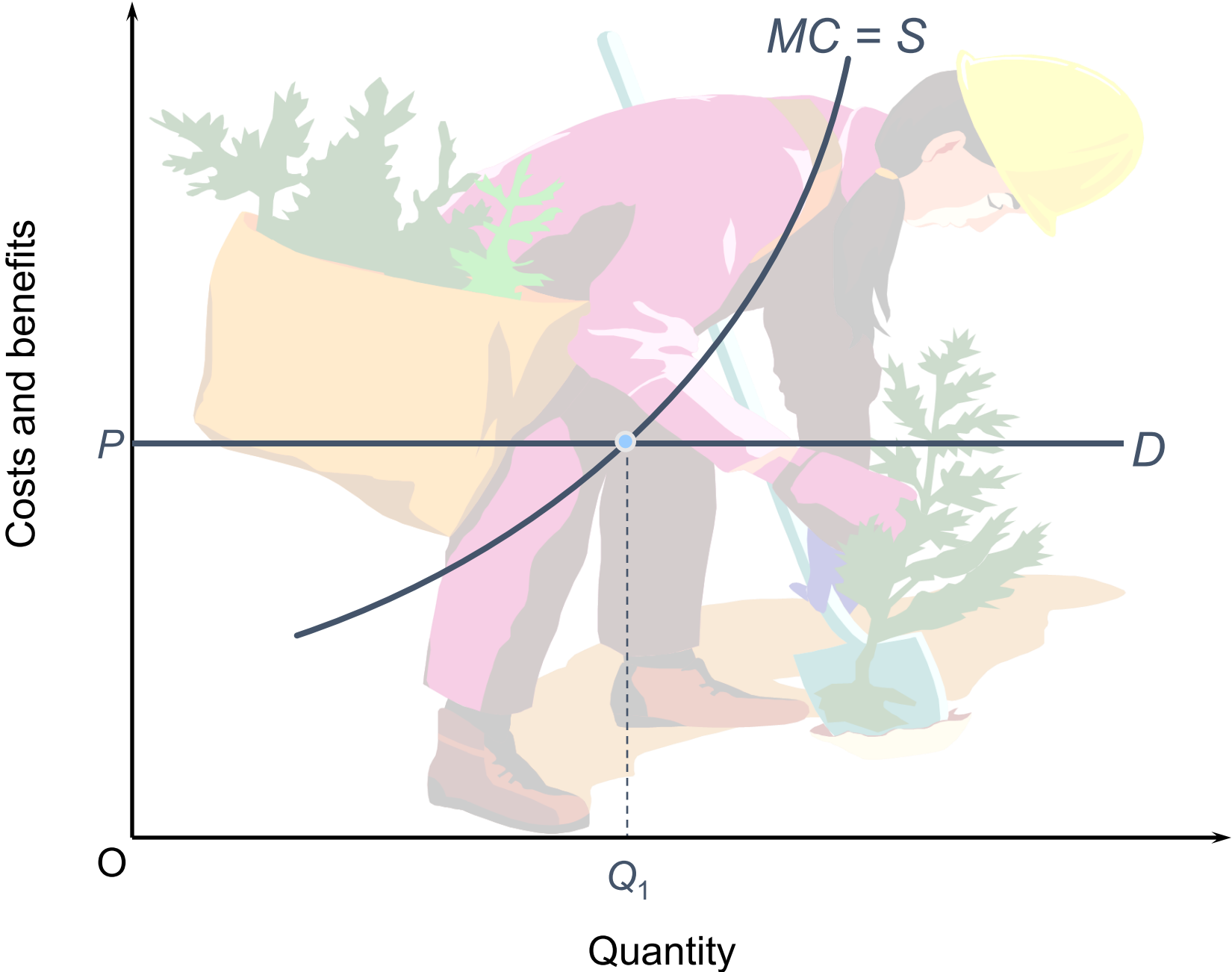
Positive Production Externalities (MSC < MPC)

- This occurs when a firm's production process creates benefits for third parties that the firm is not compensated for.
- Example: A timber company planting vast forests. Beyond selling wood, the trees reduce CO2 levels, benefiting the global population.
- $MSC = MPC - MEB_p$ (where MEB_p is the Marginal External Benefit of production).

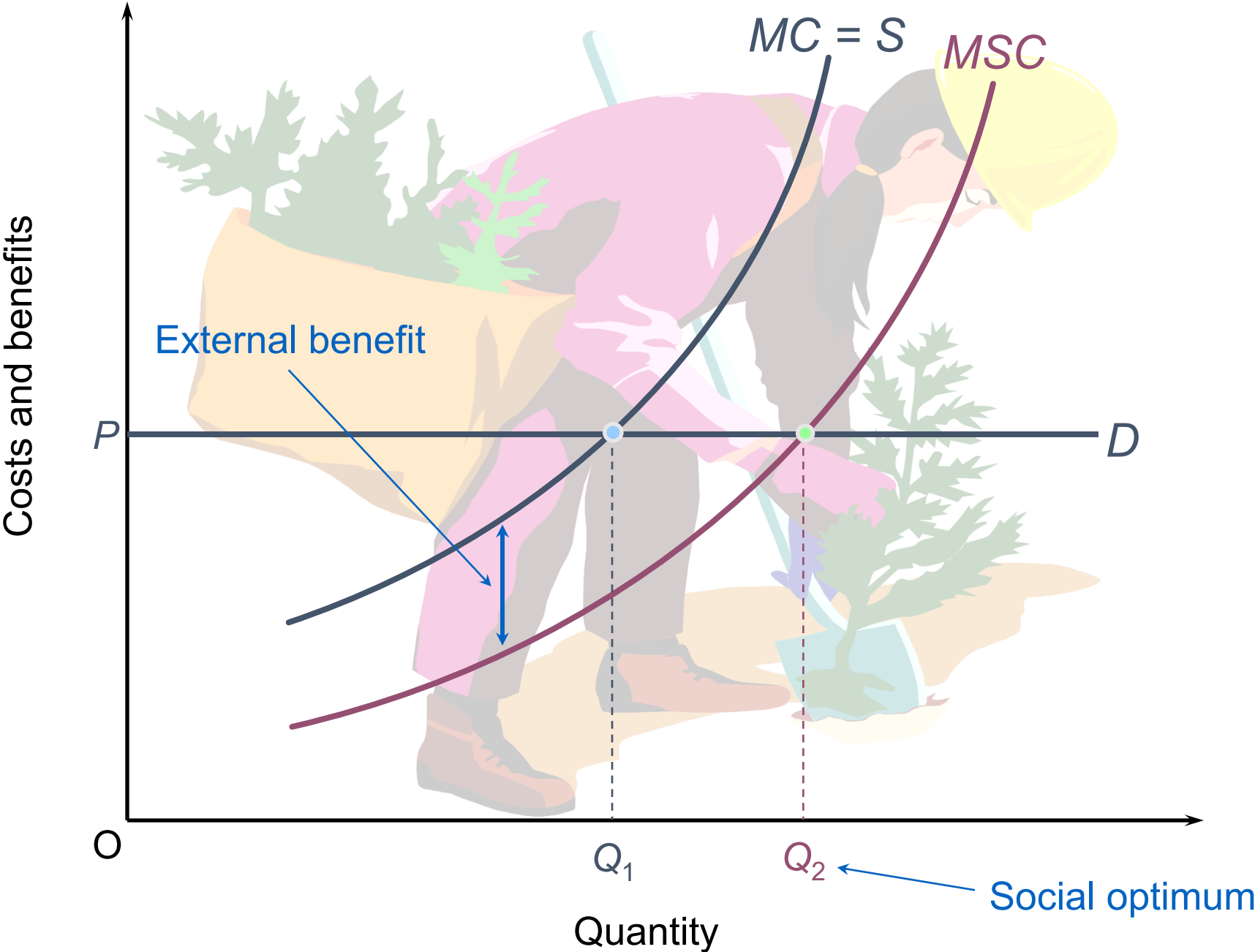
Positive Production Externalities (MSC < MPC)

- The MSC curve lies below the MPC curve. The vertical distance between them represents the external benefit to society.
- Assumptions: We assume no consumption externalities, so Demand = MSB = MPB.
- Market Outcome vs. Social Optimum:
- The Free Market: Equilibrium is at Q1 (where MPB = MPC).
- Social Efficiency: Ideally should be at Q2 (where MSB = MSC).
- Because the firm isn't paid for the external benefit, it produces less than the social optimum ($Q1 < Q2$). From society's view, not enough trees are being planted.

External benefits in production



External benefits in production

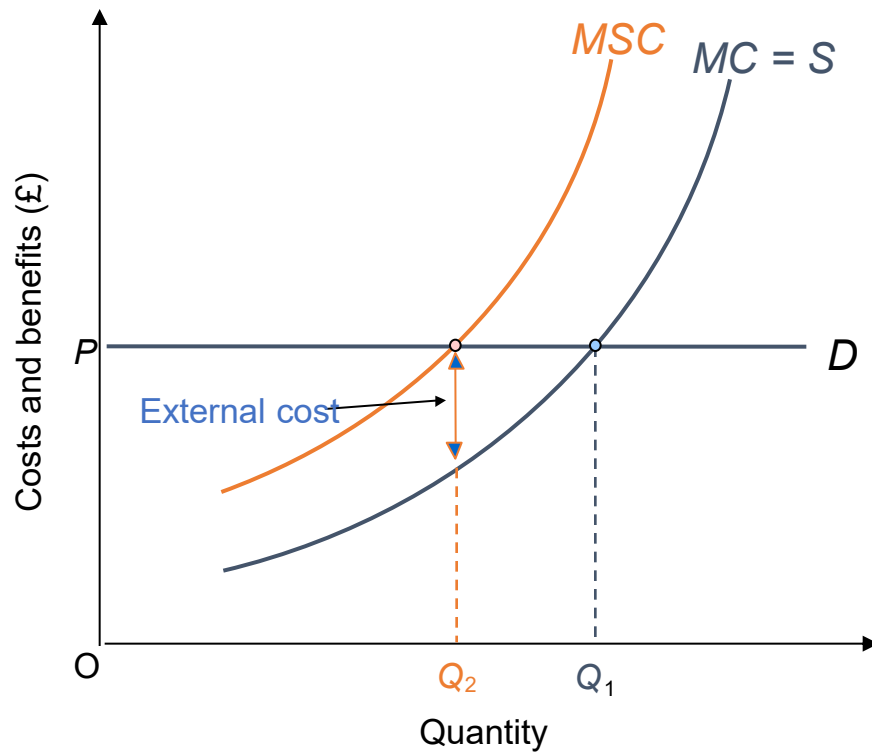


Production Externalities

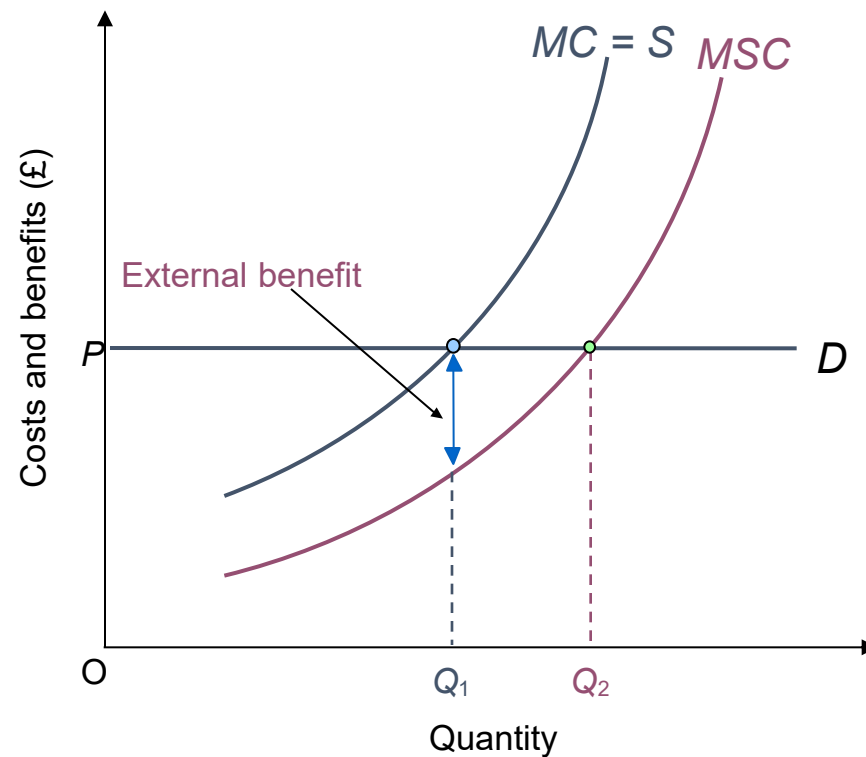
- The impact on society depends entirely on whether the externality is a cost or a benefit:

Externality Type	Cost Relationship	Market Result	Social Implication
Negative (External Cost)	$MSC > MPC$	Overproduction	Too many resources allocated ($Q1 > Q2$)
Positive (External Benefit)	$MSC < MPC$	Underproduction	Too few resources allocated ($Q1 < Q2$)

External costs and benefits in production



(a) External costs



(b) External benefits

Negative Consumption Externalities ($MSB < MPB$)

In this scenario, the individual's enjoyment of a product creates costs for others that are not reflected in the market price.

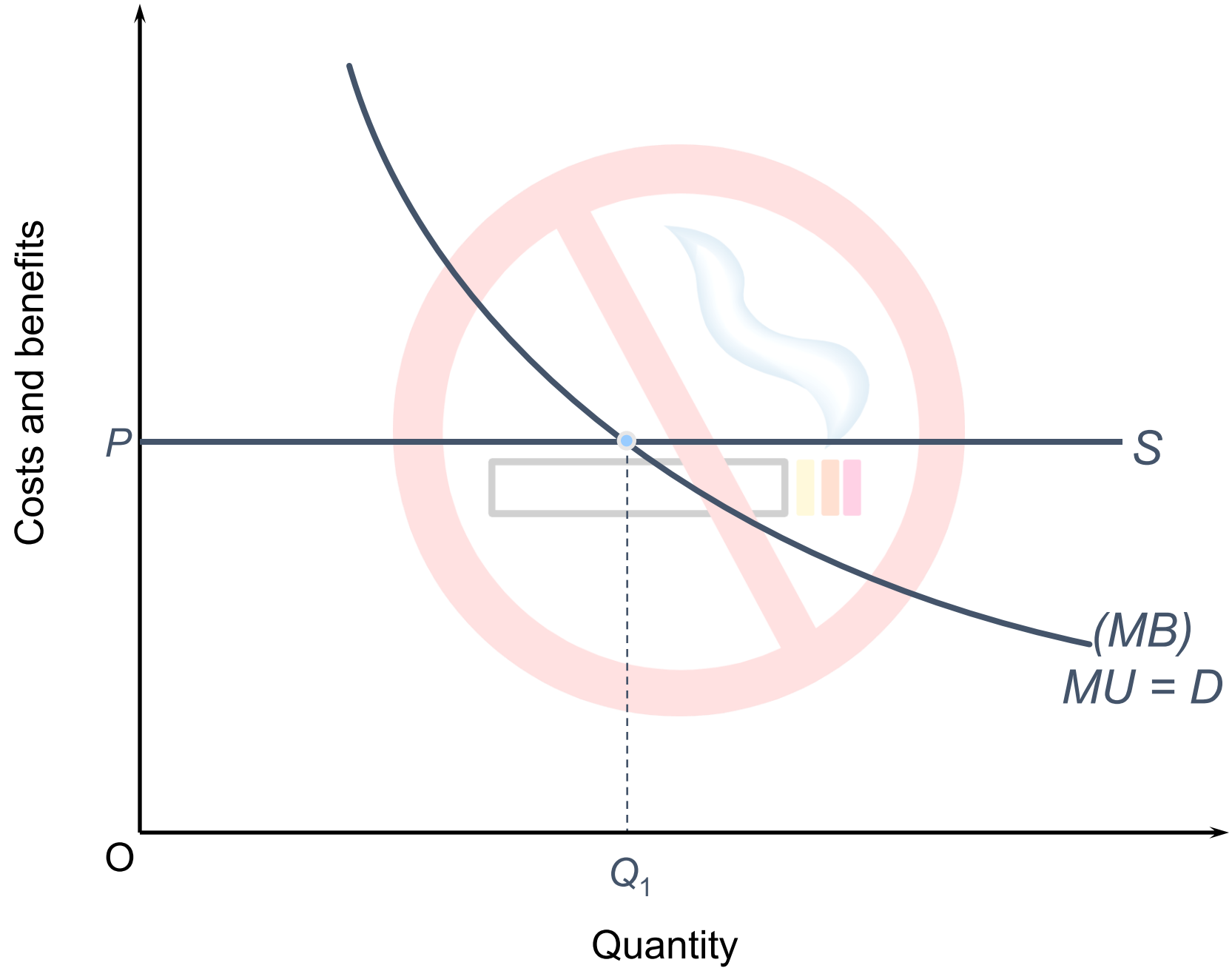
- Key Condition: Marginal Social Benefit (MSB) is less than Marginal Private Benefit (MPB).
- $MSB = MPB - MEC_c$ (where MEC_c is the Marginal External Cost of consumption).
- No Production Externalities: For this model, we assume $MSC = MPC$.

Negative Consumption Externalities ($MSB < MPB$)

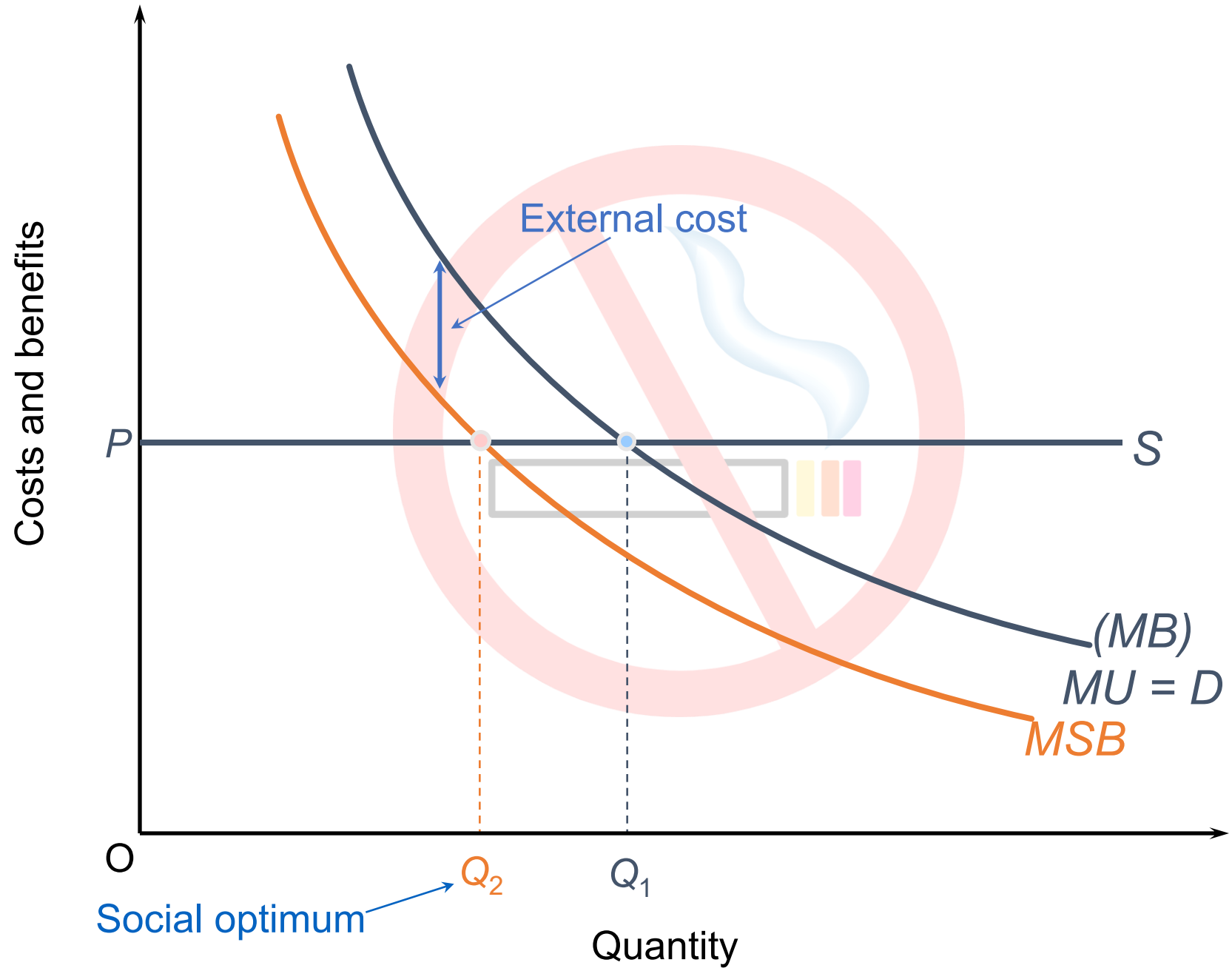
Examples: Alcohol: Private pleasure for the consumer, but results in social disorder (policing costs) and increased public health expenditures. Using a mobile phone while driving, speeding, or passive smoking.

- The MSB curve lies below the MPB (Demand) curve.
- Market Failure:
 - Free Market: Consumers choose quantity Q_1 (where $MPB = MSC$).
 - Social Optimum: Society prefers quantity Q_2 (where $MSB = MSC$).
- The market consumes too much of the good relative to what is socially desirable ($Q_1 > Q_2$).

External costs in consumption



External costs in consumption



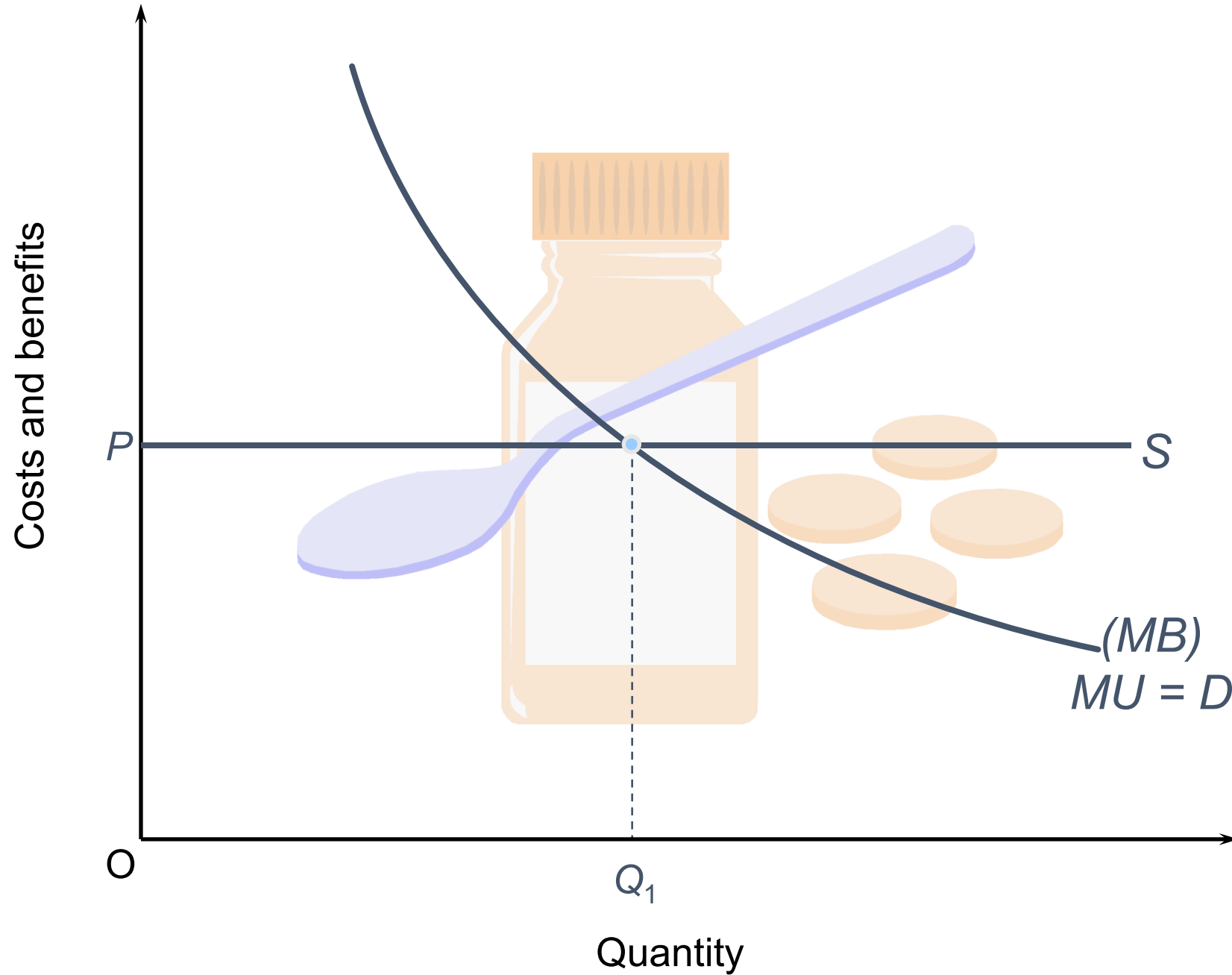
Positive Consumption Externalities ($MSB > MPB$)

- This occurs when an individual's consumption creates spillover benefits for the rest of society.
- Marginal Social Benefit (MSB) $>$ Marginal Private Benefit (MPB).
- $MSB = MPB + MEB_c$ (where MEB_c is the Marginal External Benefit of consumption).

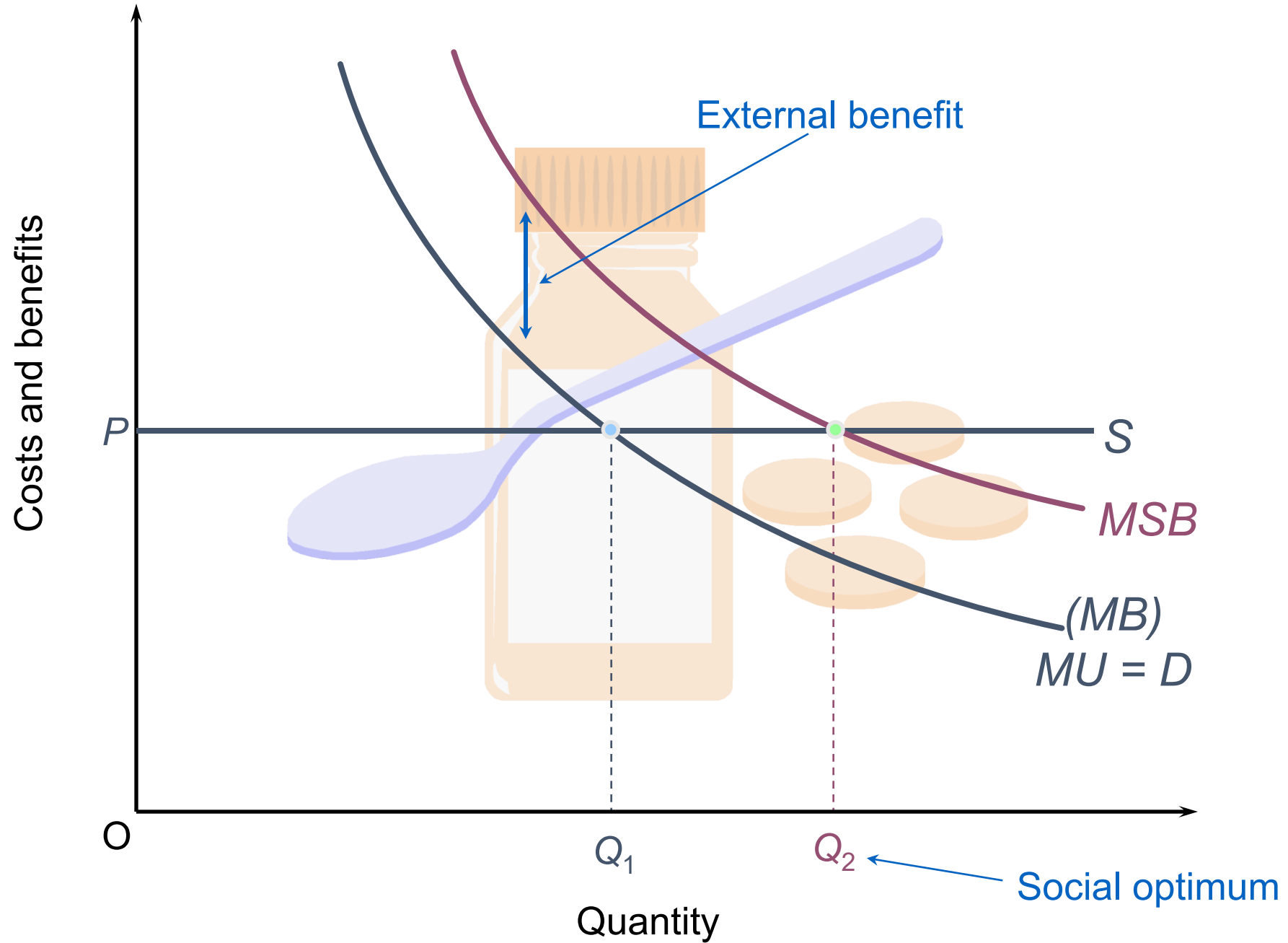
Positive Consumption Externalities ($MSB > MPB$)

- Examples: Planting flowers in a front garden provides aesthetic pleasure to the owner but also benefits neighbors and passersby. Vaccinations: Protecting yourself also prevents the spread of disease to others.
- The MSB curve lies above the MPB (Demand) curve.
- Market Failure:
 - Free Market: Equilibrium at $Q1$ (where $MPB = MSC$).
 - Social Optimum: Equilibrium at $Q2$ (where $MSB = MSC$).
- Because the consumer isn't rewarded for the external benefit, they buy less than is socially optimal ($Q1 < Q2$).

External benefits in consumption



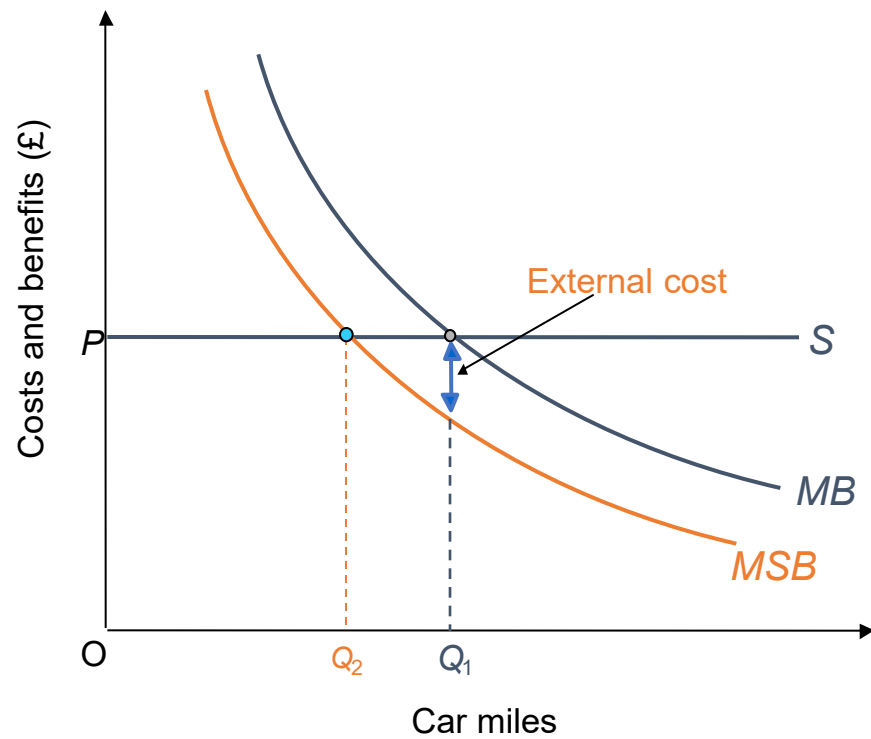
External benefits in consumption



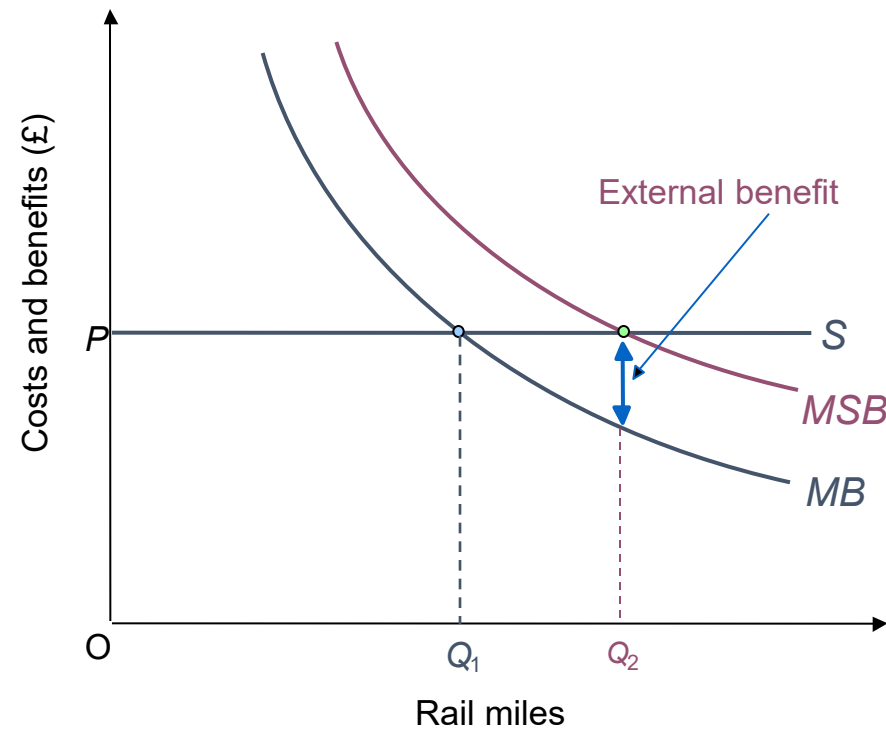
Consumption Externalities

Situation	Social vs. Private	Market Outcome
External Cost (Negative)	$\text{Social} < \text{Private}$	Too Much produced/consumed
External Benefit (Positive)	$\text{Social} > \text{Private}$	Too Little produced/consumed

External costs and benefits in consumption



(a) External costs



(b) External benefits

Government Intervention in the Market

Correcting Market Imperfections (Monopoly)

Governments use different tools to handle the inefficiencies caused by market power:

- To Tackle Excessive Profit: A Lump Sum Tax can be imposed. This is a fixed cost that does not change based on output levels or price.
- To Tackle Underproduction: If a monopolist produces too little ($Q < Q_{\text{social}}$), the government can provide a Per Unit Subsidy to lower marginal costs and encourage more output.

Correcting Negative Externalities with Taxes

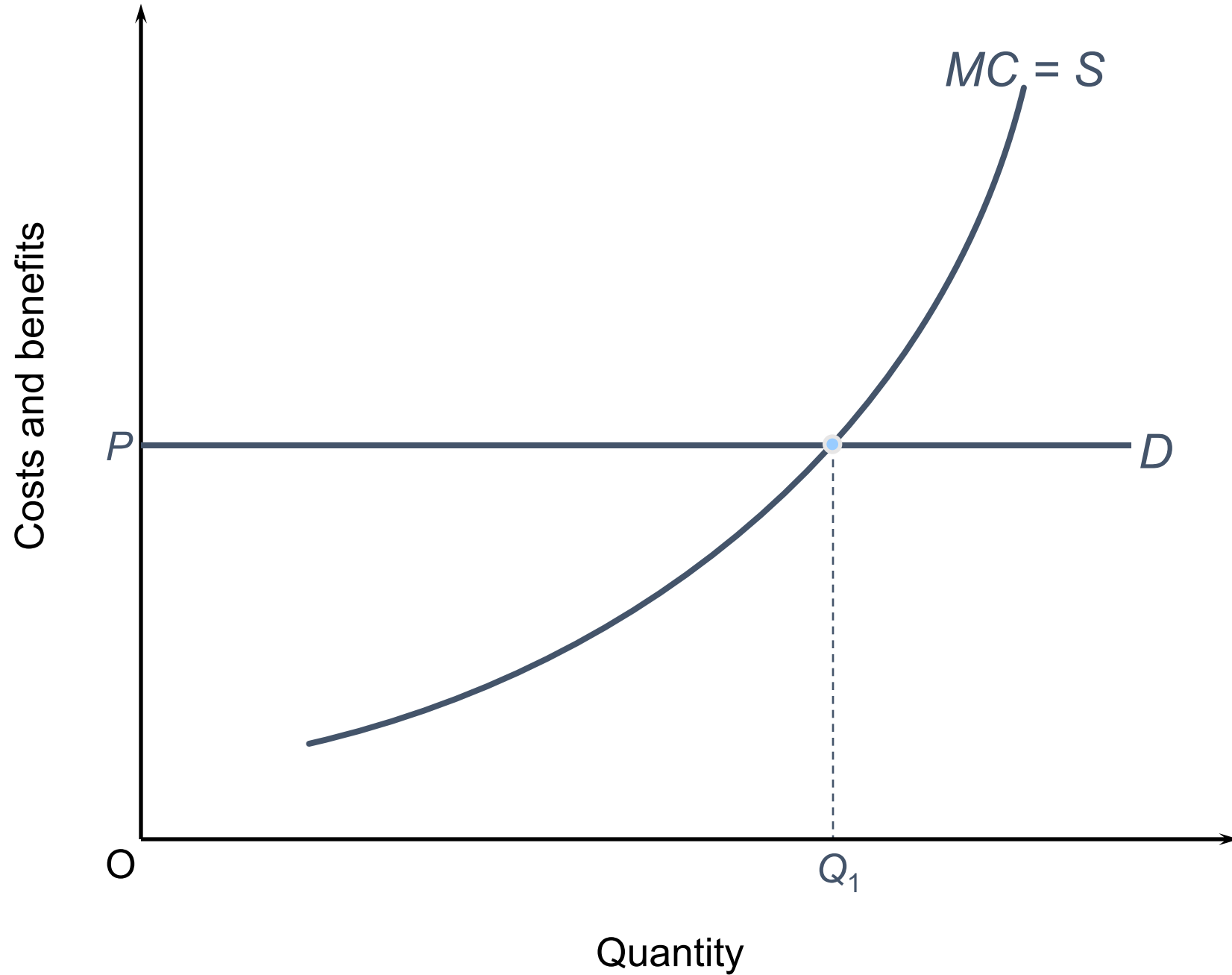


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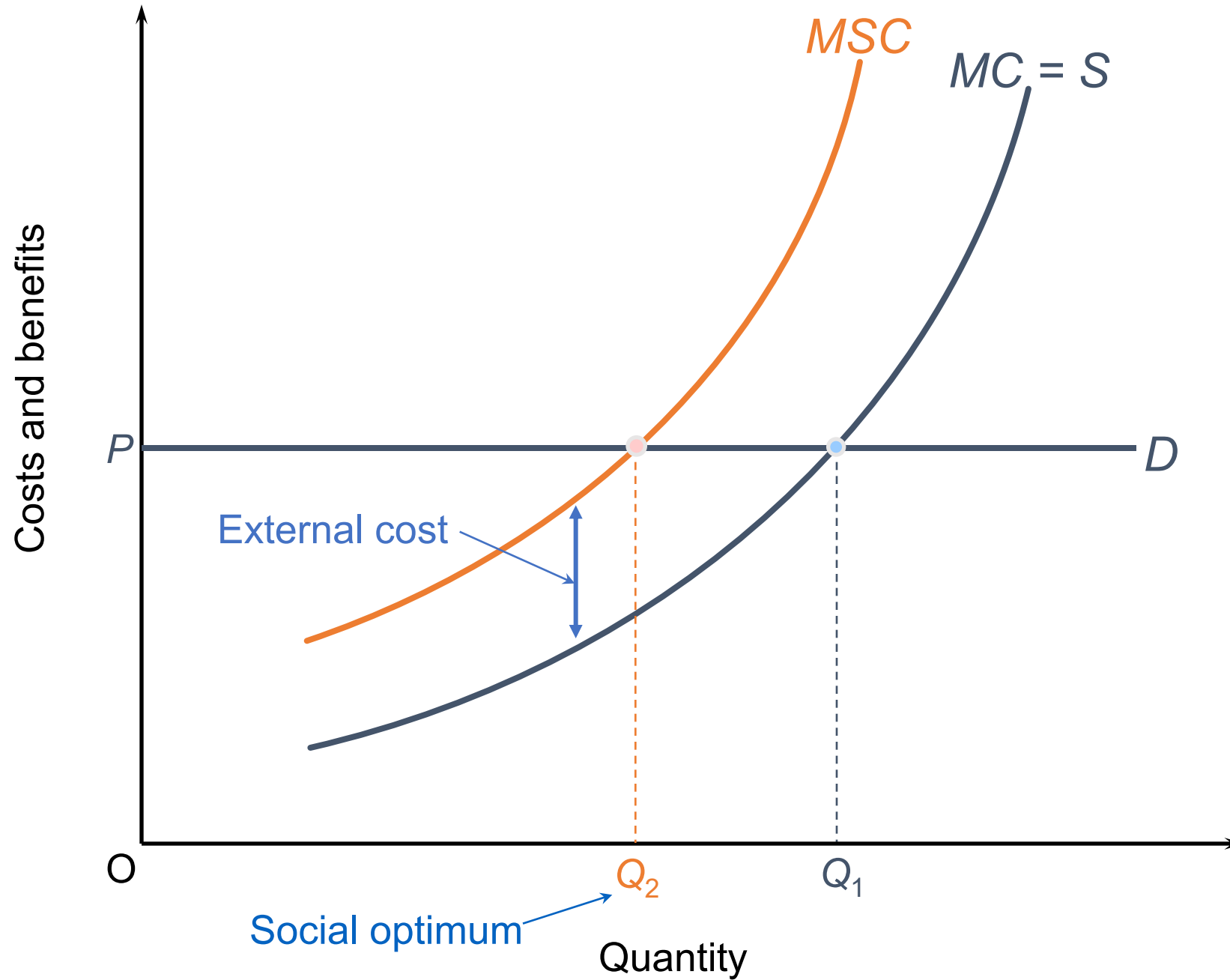
To fix overproduction/overconsumption, the government must Internalise the Externality.

- The Rule: Impose a tax exactly equal to the Marginal External Cost (MEC).
- Mechanism:
 - The tax shifts the private cost curve (MPC) upward until it aligns with the social cost curve (MSC).
 - The firm/consumer now feels the cost they were imposing on others.
- Outcome: The new profit-maximizing equilibrium shifts from Q1 to Q2 (the social optimum).
- Optimal Tax Formula: $\text{Tax} = \text{MSC} - \text{MPC}$.

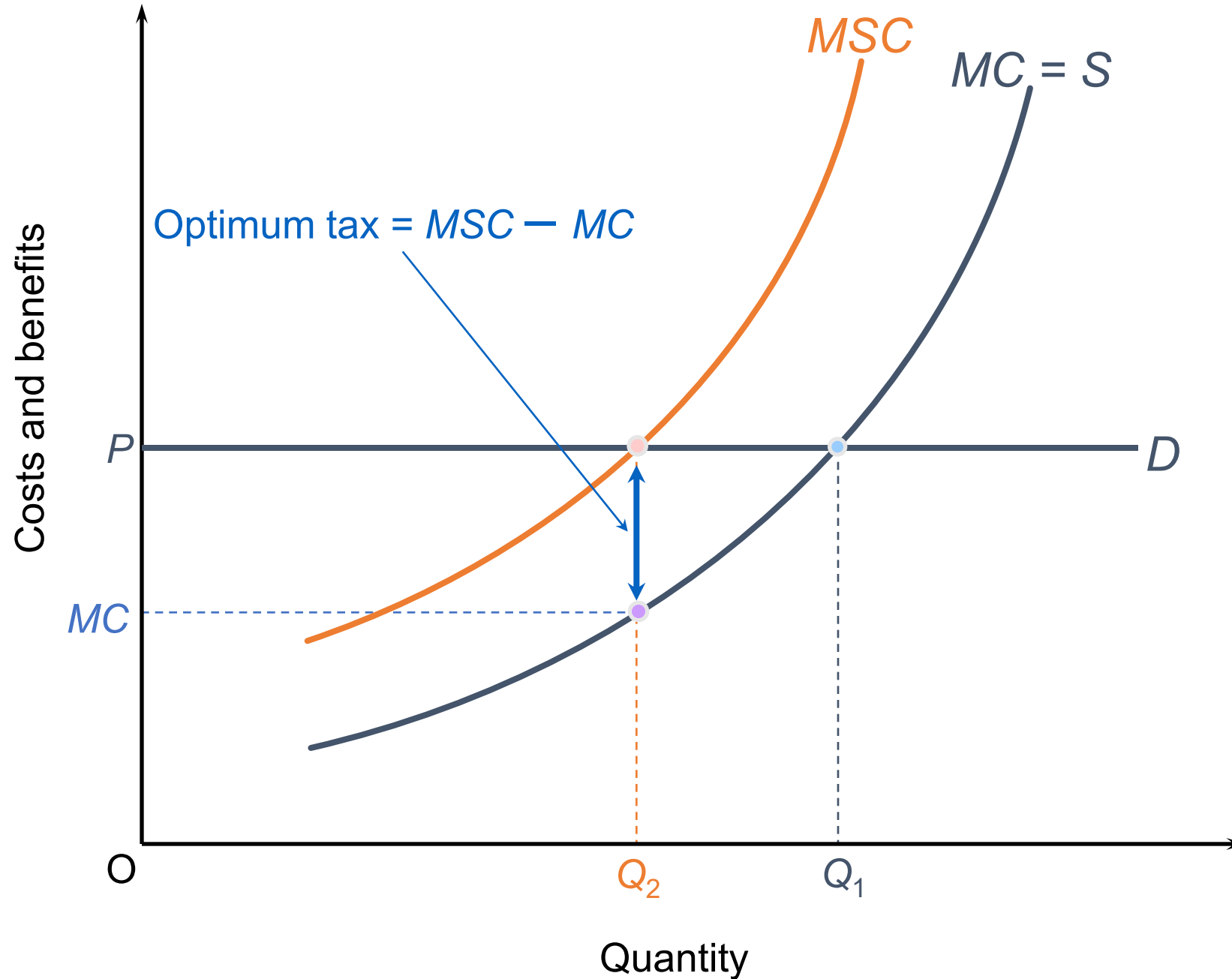
Using (Pigouvian) taxes to correct a market distortion



Using taxes to correct a market distortion



Using taxes to correct a market distortion

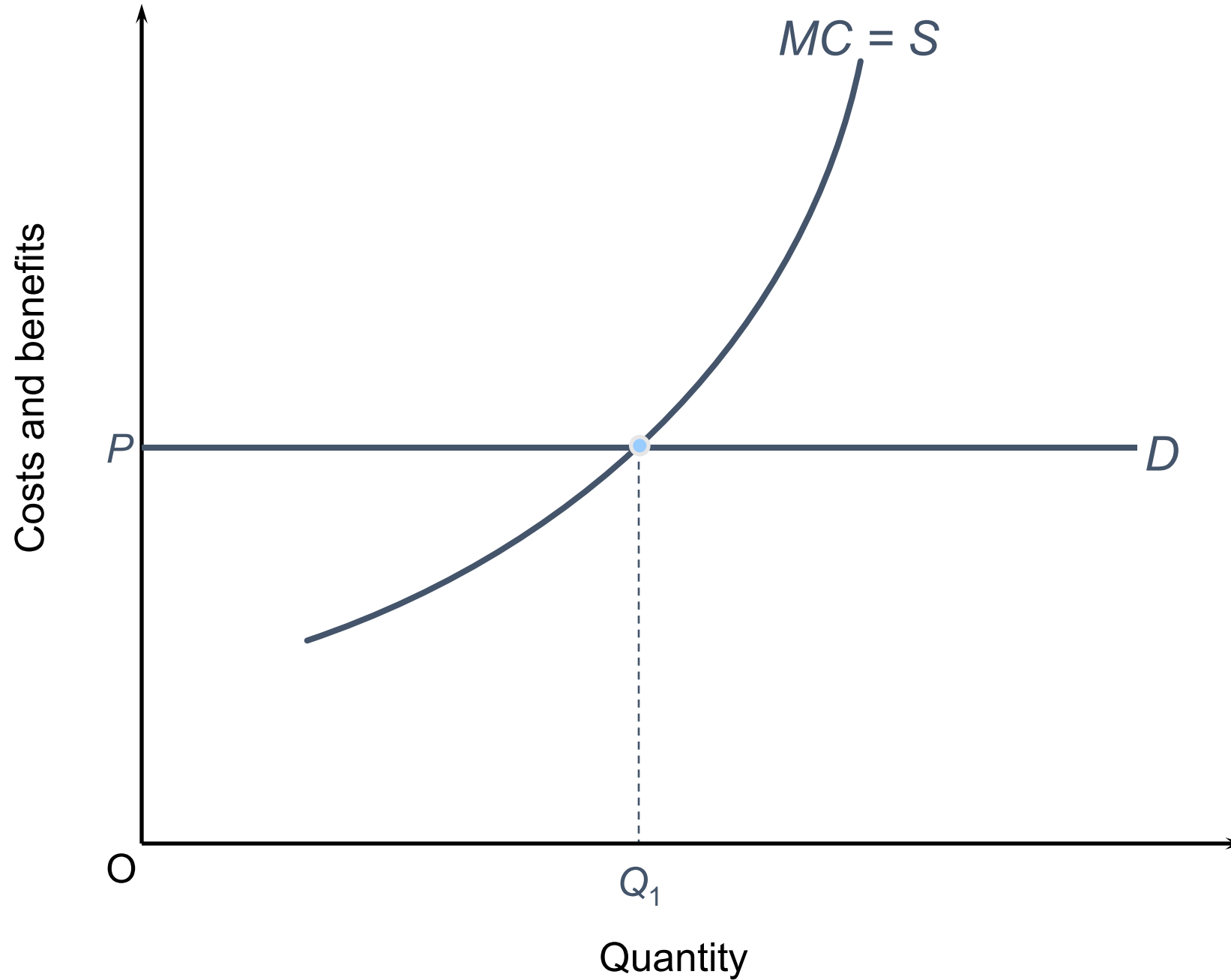


Correcting Positive Externalities with Subsidies

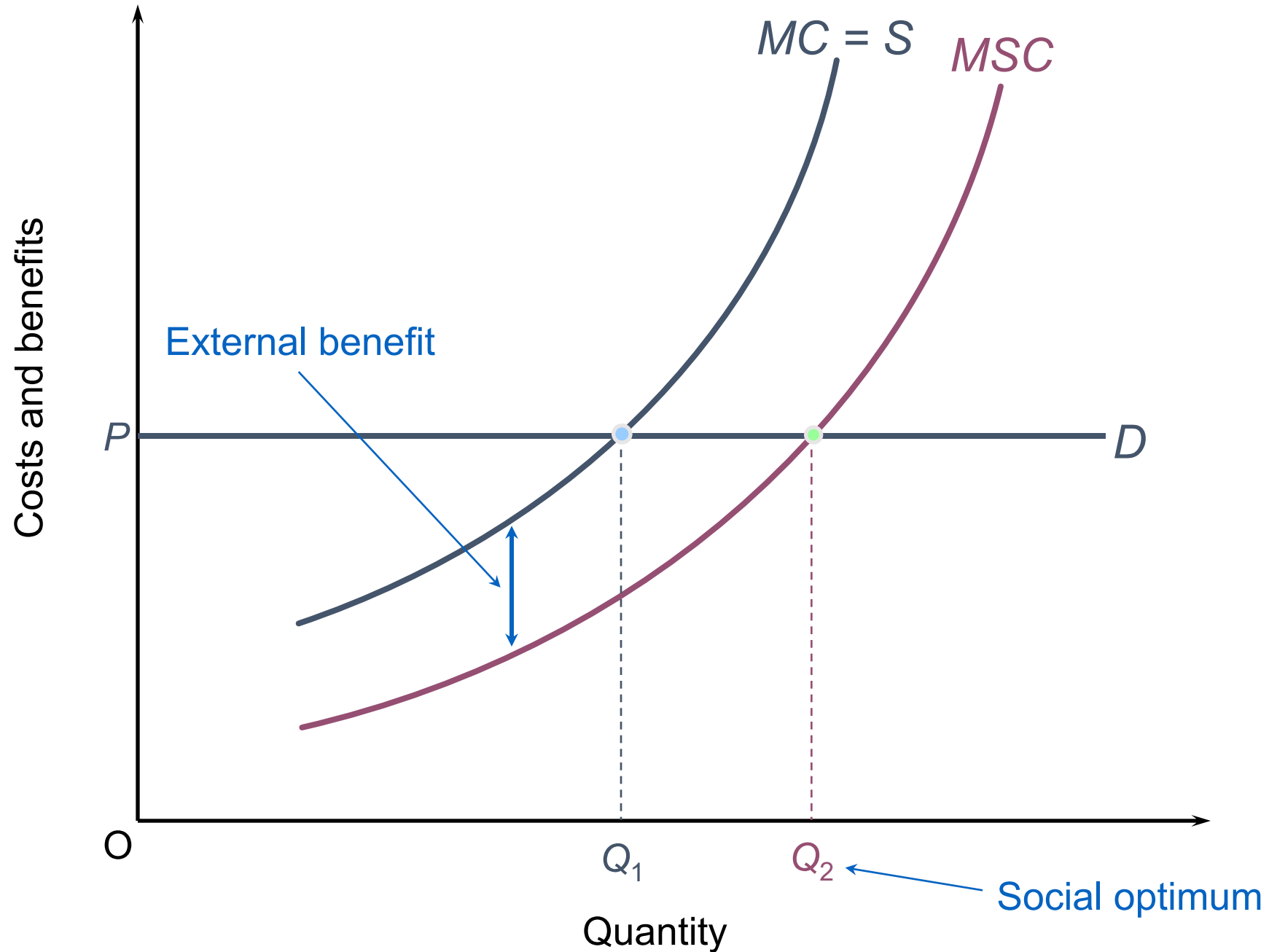
To fix underproduction/underconsumption, the government provides financial incentives.

- The Rule: Provide a subsidy exactly equal to the Marginal External Benefit (MEB).
- Mechanism:
 - The subsidy reduces the private cost to the firm (MPC), shifting the supply curve downward.
 - This encourages the firm to expand production.
- Outcome: Production increases from Q_1 to Q_2 , reaching the point where $MSB = MSC$.
- Optimal Subsidy Formula: $\text{Subsidy} = MPC - MSC$ (at the social optimum).

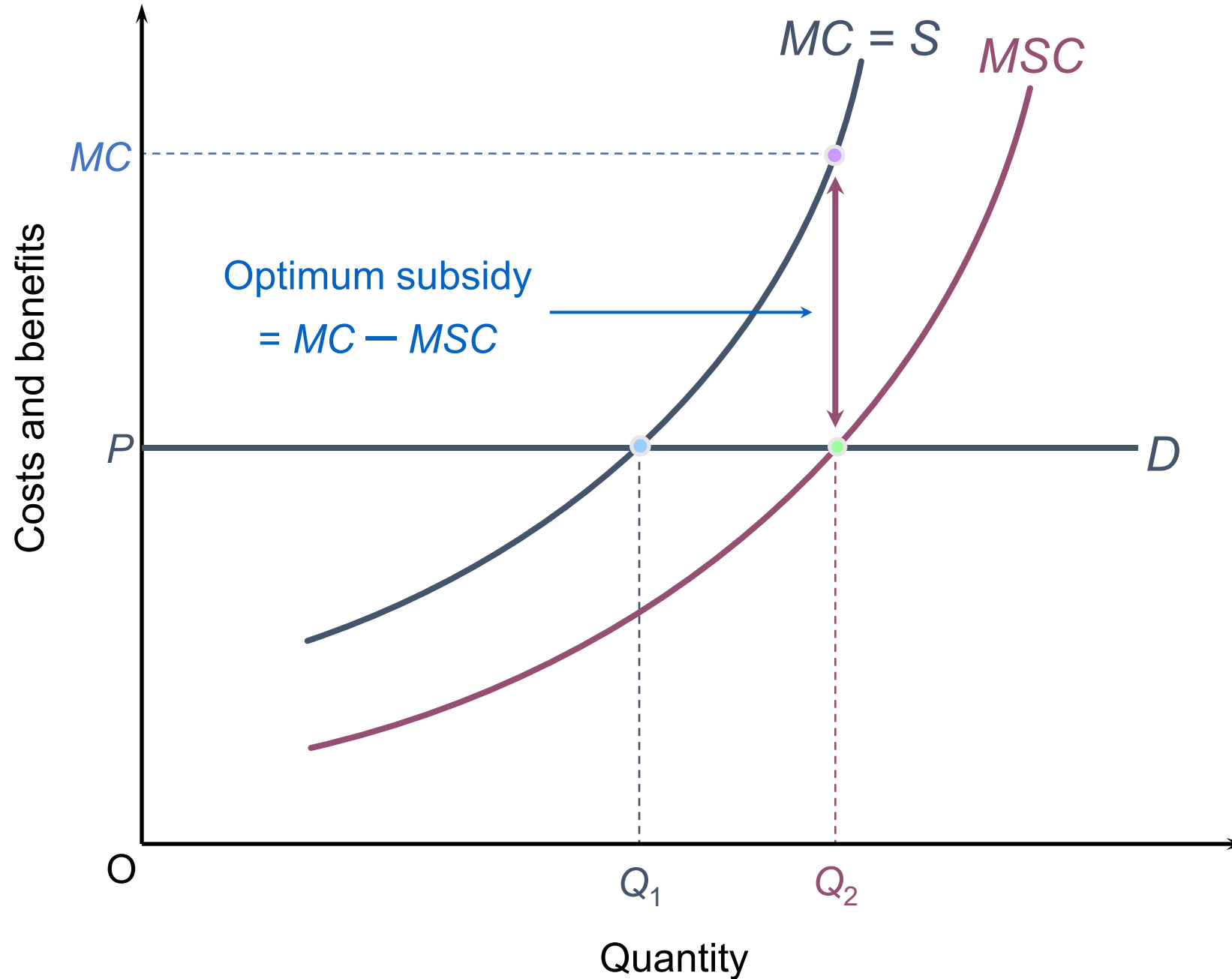
Using subsidies to correct a market distortion



Using subsidies to correct a market distortion



Using subsidies to correct a market distortion



Advantages and Disadvantages of Taxes and Subsidies

Economists often prefer market-based solutions like taxes and subsidies to correct externalities because they work within the price mechanism.

Advantages

- Market Operation: Allows the market to continue functioning while forcing firms to internalise full social costs and benefits.
- Flexibility: Tax and subsidy levels can be adjusted based on the severity of the market failure.
- Dynamic Incentives:
 - Pollution Taxes: Provide a continuous incentive for firms to innovate and find cleaner production methods to save money.
 - Subsidies: Encourage the long-term adoption of good practices by making them more profitable.

Advantages and Disadvantages of Taxes and Subsidies

Disadvantages

- **Administrative Complexity:** It is nearly impossible to calculate and apply unique tax/subsidy rates for every individual firm and its specific externalities.
- **Information Gaps (Lack of Knowledge):**
 - Difficult to measure the exact monetary value of external costs (e.g., how much is a ton of CO₂ worth?).
 - Hard to determine exactly who is to blame and how long-lasting the environmental damage will be.

Legal Restrictions and Regulation

Direct legislation (bans, quotas, or standards) offers a non-market alternative to managing undesirable economic behavior.

Advantages of Legal Restrictions

- **Simplicity and Clarity:** Usually easy for firms to understand and for authorities to administer (e.g., a simple ban on a toxic chemical).
- **Safety in Extremes:** When a risk is high (e.g., radioactive waste), a total ban is safer than relying on a tax to discourage use.
- **Emergency Speed:** Allows for rapid action during crises, such as banning cars during a smog emergency.
- **Consumer Protection:** Corrects imperfect information by making it illegal to sell unsafe goods or make false claims.

Legal Restrictions and Regulation

Disadvantages of Legal Restrictions

- **The Blunt Weapon Problem:** Unlike taxes, laws provide no incentive to improve beyond the legal limit.
 - *Example:* If the limit is 20 tons of waste, a firm has no reason to reduce it to 5 tons. A tax would reward them for every ton reduced.

Regulatory Bodies

A more sophisticated approach than general law, involving specialized agencies that investigate and manage specific sectors.

The Subtle Approach: Agencies conduct detailed investigations and issue reports or enforcement actions based on specific findings.

Key Examples:

CMA (Competition and Markets Authority): Investigates monopolies and mergers.

Ofwat / Ofgem: Regulate water and energy utilities to prevent abuse of market power.

Regulatory Bodies

Advantages:

- Uses a case-by-case method, allowing for tailored solutions rather than one-size-fits-all laws.

Disadvantages:

- Investigations are expensive and time-consuming.
- Regulatory bodies can only monitor a few cases at a time.
- **Enforcement Issues:** Firms may make promises of good behavior that are difficult to track and enforce over the long term.

Price Controls, Information, and Public Goods



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When taxes and regulations are insufficient, the government may intervene directly in pricing, information flow, or the provision of the goods themselves.

1. Price Controls

Governments can mandate prices that differ from the free-market equilibrium to protect consumers or producers.

Price Controls, Information, and Public Goods

Prices Set Below Equilibrium (Price Ceilings/Maximum Prices):

- Goal: To prevent monopolies/oligopolies from overcharging or to ensure affordability.
 - Utility Regulation: Ofgem's Safeguard Tariff (2017/2018) capped energy bills for prepayment and low-income customers due to lack of competition.

Prices Set Above Equilibrium (Price Floors/Minimum Prices):

- Goal: To reduce social inefficiency from negative externalities or to protect producer incomes.
 - Income Support: Minimum wage legislation for workers and high guaranteed farm prices for farmers.
- The Risk: Artificial price controls often lead to shortages (if the price is too low) or surpluses (if the price is too high).

Price Controls, Information, and Public Goods



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2. Provision of Information

- Market failure often stems from asymmetric information or consumer ignorance. The government acts as an information hub to improve market transparency.
- **Labour Markets:** Job centres provide information to the unemployed, increasing the elasticity of labour supply.
- **Consumer Health:** Public health campaigns regarding the dangers of smoking or poor nutrition.
- **Business Planning:** Providing national statistics on inflation, employment, and sales trends to help firms plan with more certainty.

Price Controls, Information, and Public Goods

3. Direct Provision of Goods and Services

Certain goods are not provided efficiently (or at all) by the private sector due to their nature.

- **Public Goods:** Goods that are non-excludable and non-rivalrous (e.g., national defence, flood defences).
- **The Free-Rider Problem:** Because people cannot be excluded from using these goods once provided, the market fails to produce them because no one wants to pay voluntarily.
- **Government Action:**
 - Finances these goods through compulsory taxation.
 - Manages production directly through public agencies or pays private firms to provide them on society's behalf.

The Case for Laissez-faire

The Case for Laissez-faire

The argument for a Laissez-faire approach is based on the idea of Government Failure, where intervention creates problems greater than the market failures it aims to fix.

The Case for Laissez-faire

Drawbacks of Government Intervention

- **Creation of Shortages and Surpluses:**
 - **Maximum Prices (Price Ceilings):** Setting prices below equilibrium (e.g., rent control) leads to excess demand. This necessitates waiting lists, rationing, or results in underground black markets.
 - **Minimum Prices (Price Floors):** Setting prices above equilibrium (e.g., high farm prices) leads to excess supply. The government must then store, destroy, or subsidize the disposal of the surplus.
- **Poor Information:** Governments often lack precise data on social costs and benefits, leading to policies that misinterpret consumer behavior or wishes.

The Case for Laissez-faire

Drawbacks of Government Intervention

- **Bureaucracy and Inefficiency:** Large-scale intervention requires significant administrative resources (civil servants, offices, equipment), which may be used wastefully.
- **Lack of Market Incentives:**
 - Subsidies can keep inefficient firms on life support.
 - Welfare provisions may reduce the incentive for individual effort.
- **Policy Shifts:** Frequent changes in tax rates or subsidies create an unpredictable environment, making it difficult for firms to engage in long-term planning.
- **Loss of Individual Freedom:** Intervention restricts the ability of individuals to make their own economic choices, which is often viewed as a value in itself.

The Case for Laissez-faire

Advantages of the Free Market

Proponents argue that even an imperfect market has inherent strengths that top-down regulation cannot replicate.

Efficiency and Innovation

- **Automatic Adjustments:** The market responds to changes in supply and demand automatically through the price mechanism, requiring no expensive administration or bureaucracy.
- **Dynamic Incentives:** The prospect of high profits encourages entrepreneurs to innovate.
 - New products and techniques are developed.
 - Initial high prices eventually fall as new competitors are attracted to the industry.

The Case for Laissez-faire

Competitive Pressures in Imperfect Markets

Even under Monopoly or Oligopoly, competitive behavior is often maintained due to:

- **Contestability:** The fear that high profits will attract new hit-and-run entrants.
- **Inter-industry Competition:** Substitutes from related sectors (e.g., trains vs. coaches).
- **Foreign Competition:** The threat of global rivals entering the local market.
- **Countervailing Power:** Large producers (e.g., detergent brands) must negotiate with large, powerful buyers (e.g., supermarket chains), which helps keep prices in check.
- **Corporate Control:** The threat of a takeover keeps management focused on efficiency.

Questions